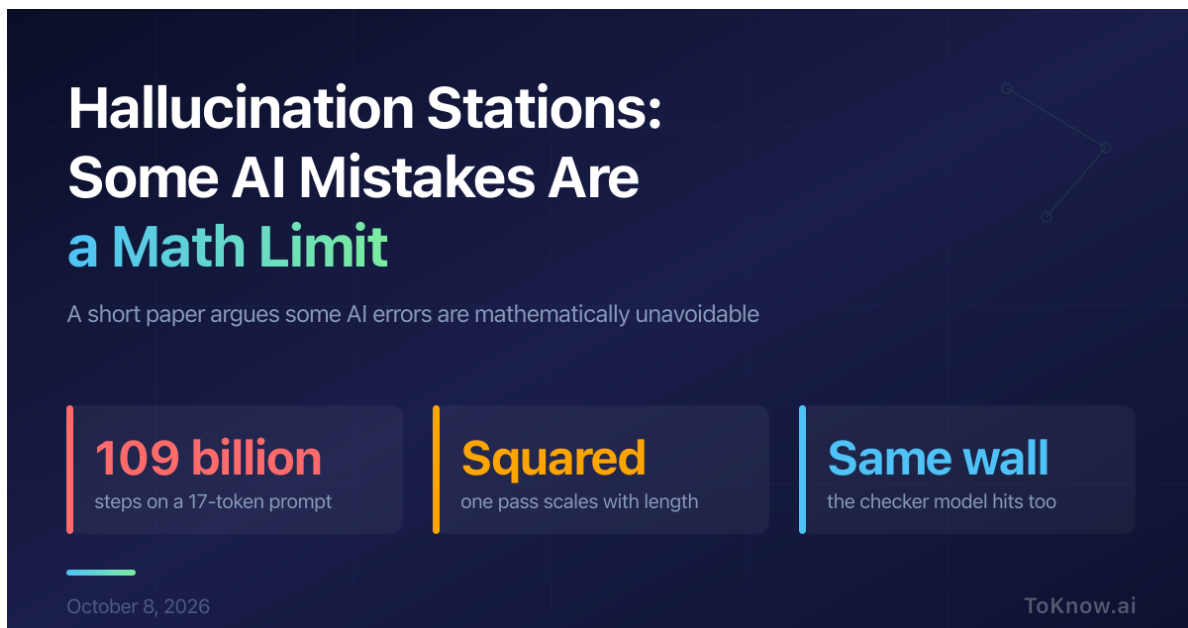


Hallucination Stations: Some AI Mistakes Are a Math Limit, Not a Bug

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**Hallucination Stations:
Some AI Mistakes Are
a Math Limit**

A short paper argues some AI errors are mathematically unavoidable

- 109 billion**
steps on a 17-token prompt
- Squared**
one pass scales with length
- Same wall**
the checker model hits too

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Two researchers argue that some AI mistakes are not bugs but a hard ceiling on how much arithmetic a language model can do in one pass. [Hallucination Stations](#) is a short paper by Varin Sikka, a Stanford student, and his father Vishal Sikka, a former Infosys chief executive. The claim comes from how a model works: each pass costs about the square of the input length, so every prompt gets the same fixed computation. They measured one. A small Llama

model runs exactly 109,243,372,873 floating-point operations, the add-and-multiply steps a chip performs, on any 17-token prompt, whether it explains renewable energy or solves a puzzle. When a task needs more steps than that budget allows, the model cannot get it right.

Listing every combination of a set of tokens, or multiplying two large matrices, needs far more steps than one pass supplies. The paper states it as a theorem: if a prompt contains a task more complex than the model's fixed budget, the answer will be wrong, following a [1965 result](#) that says some problems need more steps than others. That matters for scheduling, routing, software verification and simulation, where teams now hand an agent a problem and trust the number it returns.

Checking the work does not help: a second model hits the same ceiling, and verifying a hard answer can cost more than producing it. The bound covers one uninterrupted guess. A model that writes out a long [chain of thought](#) or runs an offline review pass spends more computation and can reach further, though the authors say those extra tokens are far fewer than the task needs.

Read More: [Do AI Models Need Sleep? An Offline Pass That Boosts Reasoning.](#)

Sources:

- [Hallucination Stations: On Some Basic Limitations of Transformer-Based Language Models \(arXiv\)](#)
- [AI Agents Are Poised to Hit a Mathematical Wall, Study Finds \(Gizmodo\)](#)
- [The Math on AI Agents Does Not Add Up \(WIRED\)](#)
- [The Illusion of Thinking \(Apple Machine Learning Research\)](#)
- [Chain of Thought Empowers Transformers to Solve Inherently Serial Problems \(arXiv\)](#)

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